

# Split Complex Lorentzian Dynamics

Part XXVIII: Continuous Internal Representatives, Kernel Defects, and the Global Compatibility Frontier

From continuous ordinary transition data to locally constant multiple-overlap defects, with simultaneous compatibility isolated for Part XXIX

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## Abstract

Part XXVII closed the ordinary topological side of the reconstruction. Starting from varying metric-oriented real three-spaces over a topological base, a compatible local atlas was promoted to a uniquely characterized total-space topology. The oriented frame total space inherited genuine local product charts, and the ordinary overlap maps became continuous functions

$$g_{ij} : U_i \cap U_j \longrightarrow G_{\text{vis}}$$

satisfying the exact laws

$$g_{ii} = 1, \quad g_{ji} = g_{ij}^{-1}, \quad g_{ik} = g_{jk}g_{ij}.$$

Separately, earlier parts had already reconstructed a topological group  $\mathcal{L}$ , a continuous surjective group homomorphism  $p : \mathcal{L} \rightarrow G_{\text{vis}}$ , local continuous sections of  $p$ , and for the certified model the exact kernel  $K = \{\pm 1\}$  [7, 6, 1]. The outstanding question was therefore no longer whether either side existed. It was whether the two independently reconstructed sides could be made locally and then globally compatible.

Task 28 attacks precisely this interface, without importing a principal bundle, a named lifted global structure, a characteristic class, or cohomology. It first abstracts the already certified internal projection into an **InternalProjection** interface: a continuous surjective homomorphism with local continuous sections, central kernel, and discrete kernel. For any continuous ordinary map  $g : X \rightarrow G_{\text{vis}}$ , every point  $x \in X$  is proved to admit an open neighbourhood  $V$  and a continuous internal representative  $u : V \rightarrow \mathcal{L}$  with  $p \circ u = g$  on  $V$ . Applied to the ordinary transition maps, this yields open refinements of every overlap carrying continuous internal representatives. Whole-overlap representability is not inferred; for the certified projection it is explicitly refuted in general by the inherited no-global-section theorem.

The first new rigidity theorem classifies all ambiguity. If  $u$  and  $v$  continuously represent the same ordinary map on  $V$ , their relative factor

$$\varepsilon := v u^{-1}$$

is continuous and kernel-valued, is the unique function satisfying  $v = \varepsilon u$ , and conversely every continuous kernel-valued  $\varepsilon$  produces another representative. Because the kernel is discrete,  $\varepsilon$  is locally constant, and on connected domains it is constant. In the certified instance this is exactly a locally constant  $\{\pm 1\}$ -valued ambiguity; a disconnected-domain counterexample proves that “one sign per whole overlap” is generally false.

The central construction appears on refined triple overlaps. Using the inherited ordinary convention  $g_{ik} = g_{jk}g_{ij}$ , Task 28 defines

$$\delta_{ijk} := (u_{jk}u_{ij})u_{ik}^{-1},$$

not as an external obstruction object but as the relative factor between two internal representatives of the same ordinary map  $g_{ik}$ . It is proved kernel-valued, continuous, locally constant, and unique in the identity

$$u_{jk}u_{ij} = \delta_{ijk}u_{ik}.$$

Exact internal transition compatibility holds if and only if  $\delta_{ijk} = 1$ . Changing representatives by  $u'_{ij} = \varepsilon_{ij}u_{ij}$  changes the defect by the formally derived law

$$\delta'_{ijk} = (\varepsilon_{jk}\varepsilon_{ij}\varepsilon_{ik}^{-1})\delta_{ijk}.$$

Associativity in  $\mathcal{L}$  further yields the quadruple-overlap consistency relation

$$\delta_{ijk}\delta_{ikl} = \delta_{jkl}\delta_{ijl}.$$

No external obstruction formalism is used to obtain either identity.

The formal run comprises nine new Lean files including the endpoint module, 2023 new Lean lines, a focused green build, and a whole-project green build at 8317 jobs with zero errors. The new layer has zero warnings; the 114 whole-project warnings are inherited. Forty-eight audited declarations depend only on the standard baseline `propext`, `Classical.choice`, and `Quot.sound`. There are no `sorry`, `admit`, custom axioms, or prohibited proof escapes [2]. Twelve preserved failed builds are all proof-engineering or elaboration failures and change no mathematical statement.

The result substantially narrows the distance to a global lifted frame geometry while exposing its genuine difficulty. Level A, pointwise representability, was inherited. Level B, local continuous representability after refinement, is now proved. Level C, simultaneous exact compatibility, is not proved. Task 28 isolates the precise next question as `CanTrivialiseSimultaneously`: can the chosen local representatives be modified by continuous kernel-valued functions so that all triple defects become trivial at once? A review boundary is essential here: the current formal predicate `IsCompatibleInternalTransitionSystem` expresses vanishing triple defects on the refined candidate, but Task 28 does not yet prove all normalization and gluing consequences required to identify that predicate with a single globally glued transition system on the original overlaps. Part XXIX should therefore first harden the meaning of “defect-free”, then attack the simultaneous-trivialization problem intrinsically. Only after that would it be mathematically legitimate to construct a global lifted total space and compare the resulting object with the established definition of a spin structure.

**Keywords:** continuous transition functions; local internal representatives; discrete central kernel;  $\mathbb{Z}_2$  ambiguity; triple-overlap defect; quadruple-overlap consistency; local constancy; global compatibility; formal verification; spin frontier; Conservation of Difficulty.

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# 1 The first genuinely global compatibility problem

The previous stages progressively removed distinct kinds of uncertainty. The internal two-sheeted group geometry was reconstructed first; later, the ordinary oriented-frame carrier and its lifted one-fibre partner were separated from reference choices; then a varying family of metric-oriented fibres was introduced; finally, Part XXVII supplied the missing topology of variation and produced continuous ordinary transition functions. At the beginning of Part XXVIII, both sides of the prospective comparison therefore exist independently:

$$\begin{array}{ll} \text{ordinary topological frame side} & \text{internal one-fibre side} \\ g_{ij} : U_i \cap U_j \rightarrow G_{\text{vis}} & p : \mathcal{L} \rightarrow G_{\text{vis}}, \quad \ker p = \{\pm 1\}. \end{array}$$

The difficulty is no longer to discover either object. It is to determine whether continuous ordinary gluing can be represented coherently by the internal system.

A tempting but invalid shortcut would be to use surjectivity of  $p$  pointwise: for every  $b$ , choose one  $u_{ij}(b)$  with  $p(u_{ij}(b)) = g_{ij}(b)$ . Such a choice does not imply continuity. Task 28 therefore begins one level lower than global gluing and asks for local continuous representatives only. The resulting answer is simultaneously reassuring and restrictive: local representatives always exist after refinement, but the comparison of different representatives produces a discrete kernel-valued freedom, and their products on triple overlaps need agree only up to a new locally constant factor.

**Methodological principle 1** (Local solvability does not remove global difficulty). Surjectivity settles pointwise existence. Local sections settle local continuous existence. Neither statement settles compatibility of choices on multiple overlaps. The remaining difficulty is therefore not a failure of local representation but a failure of simultaneous normalization.

The central achievement of Part XXVIII is that this remaining difficulty is no longer vague. It is represented by explicit, formally derived functions  $\delta_{ijk}$  with exact algebraic and topological laws.

## 2 Formal source, scope, and provenance

The supplied Task-28 archive has SHA-256

1794051d0927bfd03466577f28460f8a8e9dbf13a1c62aae572380bcefabc40a.

The pinned environment is Lean 4.28.0 with Mathlib commit

8f9d9cff6bd728b17a24e163c9402775d9e6a365.

The Task-27 source is inherited unchanged. The new source layer is

```
TransitionLiftBase → LocalInternalRepresentatives → KernelAmbiguity →
TripleOverlapDefect → RepresentativeChange → CertifiedProjection →
InternalTransitionInterface → ModelBridge → Task28.
```

The audit prose calls this “8 new modules”, while the directory contains eight substantive support modules plus the endpoint `Task28.lean`, hence nine new Lean files in total. This is a documentation-counting convention, not a mathematical discrepancy.

| Check                 | Result | Detail                                                       |
|-----------------------|--------|--------------------------------------------------------------|
| Focused Task-28 build | PASS   | <code>lakebuildRequestProject.NullSectorTask28.Task28</code> |
| Whole-project build   | PASS   | 8317 jobs, 0 errors                                          |

| Check                    | Result          | Detail                                                                                                                                                           |
|--------------------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Task-28 warnings         | 0               | 114 project-wide warnings are inherited                                                                                                                          |
| New source               | 9 Lean files    | 2023 Lean lines                                                                                                                                                  |
| Axiom audit              | 48 declarations | each reports only <code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>                                                                 |
| <code>sorry/admit</code> | 0/0             | no proof holes                                                                                                                                                   |
| Prohibited escapes       | 0               | no custom <code>axiom</code> , <code>implemented_by</code> , <code>native_decide</code> , <code>bv_decide</code> , <code>unsafe</code> , or <code>partial</code> |
| Preserved failed builds  | 12              | all elaboration/proof-engineering failures; no theorem changed                                                                                                   |

**Proof boundary 1** (Verification boundary). *The supplied source, audit, endpoint statements, source dependencies, proof-escape scan, and the critical formulas used in this paper were inspected directly. The reported Lean builds and `#print axioms` outputs are taken from the supplied artifact; the paper-generation environment did not independently rebuild the Lean project.*

Task 28 does not construct a lifted total space, a principal internal-group object, a named global lifted structure, a characteristic class, or a spin structure. It also does not import homotopy groups, Čech cohomology, sheaf machinery, smoothness, connections, or curvature. These absences are part of the theorem boundary, not omissions to be silently filled by interpretation.

### 3 Frozen inputs: ordinary transitions and the internal projection

#### 3.1 The ordinary transition system

From Task 27, let  $B$  be the topological base with open cover  $\{U_i\}_{i \in I}$ . The ordinary frame atlas provides continuous maps

$$g_{ij} : U_i \cap U_j \longrightarrow G_{\text{vis}}$$

obeying

$$g_{ii} = 1, \quad g_{ji} = g_{ij}^{-1}, \quad g_{ik} = g_{jk}g_{ij} \quad (1)$$

on the corresponding intersections. Task 28 repackages these data as `TransitionSystem`; no new ordinary geometry is introduced.

#### 3.2 The internal projection

The one-fibre work supplies a topological group  $\mathcal{L}$  and a continuous surjective homomorphism

$$p : \mathcal{L} \longrightarrow G_{\text{vis}}.$$

Task 28 freezes only the properties required for the new arguments into `InternalProjection`:

- (a)  $p$  is a continuous group homomorphism;
- (b)  $p$  is surjective;
- (c) every  $g \in G_{\text{vis}}$  has an open neighbourhood admitting a continuous section of  $p$ ;
- (d) the kernel  $K := \ker p$  is central in  $\mathcal{L}$ ;
- (e)  $K$  is discrete.

The abstract theory in the middle of Task 28 needs only a discrete central kernel. The certified project instance is stronger:

$$K = \{\pm 1\}.$$

Keeping these two levels separate prevents the two-valued conclusion from being smuggled into the abstract proofs.

**Methodological principle 2** (Abstract first, certified specialization second). Kernel-valuedness and local constancy are proved for an arbitrary discrete central kernel. Only after those theorems are established is the certified identity  $K = \{\pm 1\}$  applied. Thus the sign structure is a derived specialization rather than a hidden assumption.

## 4 Hard target A: local continuous representability

Let  $X$  be a topological space and  $g : X \rightarrow G_{\text{vis}}$  continuous. The key local theorem is `InternalProjection.exists_local_continuous_internal_rep`.

**Theorem 1** (Local continuous representability). *For every  $x \in X$  there exists an open neighbourhood  $V \ni x$  and a continuous map  $u : V \rightarrow \mathcal{L}$  such that*

$$p(u(y)) = g(y) \quad \text{for all } y \in V.$$

The proof route is exactly the intended one. A local section  $s$  of  $p$  exists near  $g(x)$ . By continuity of  $g$ , the preimage of the section domain is an open neighbourhood of  $x$ . The local representative is then  $u = s \circ g$  on that neighbourhood. No discontinuous pointwise choice is used.

Applying theorem 1 to  $g_{ij}$  yields an open cover of each ordinary overlap  $U_i \cap U_j$  by subdomains  $V_{ij\alpha}$  with continuous internal representatives

$$u_{ij\alpha} : V_{ij\alpha} \longrightarrow \mathcal{L}, \quad p \circ u_{ij\alpha} = g_{ij}|_{V_{ij\alpha}}.$$

This is packaged by `OverlapRefinement` and, for the full transition system, by `InternalTransitionCandidate`.

### 4.1 Why refinement is mathematically necessary

The theorem is local. Task 28 refuses to promote it to a single representative on the entire original overlap. This restraint is formalized by `Certified.no_whole_domain_internal_rep`, which specializes the inherited no-global-section theorem to show that whole-domain representability is not automatic for the certified projection.

**Methodological caveat 1** (Three distinct levels of representability). *The project now separates:*

$$\begin{aligned} \text{pointwise representability} & : \forall x \exists u_x, \\ \text{locally continuous representability} & : \forall x \exists V_x, u_x(\cdot), \\ \text{whole-domain continuous representability} & : \exists u : X \rightarrow \mathcal{L} \text{ continuous.} \end{aligned}$$

*Only the first follows from surjectivity, and only the second follows from local sections. The third is an additional global condition.*

This distinction is the first instance of the new Conservation-of-Difficulty phenomenon: local solvability is inexpensive once the projection is understood, but extending a choice over a larger topology can fail.

## 5 Hard target B: exact classification of representative ambiguity

Suppose  $u, v : X \rightarrow \mathcal{L}$  are continuous on a common domain  $V$  and represent the same ordinary map  $g$  there. Task 28 defines the left relative factor

$$\varepsilon(x) := v(x)u(x)^{-1}. \quad (2)$$

Because  $p(v) = p(u)$ , homomorphicity gives

$$p(\varepsilon) = p(v)p(u)^{-1} = 1,$$

so  $\varepsilon(x) \in K$ .

The formal result is considerably stronger than kernel membership.

**Theorem 2** (Unique kernel ambiguity). *If  $u$  and  $v$  continuously represent the same ordinary map on  $V$ , then:*

- (i)  $\varepsilon = vu^{-1}$  is continuous and kernel-valued;
- (ii)  $v = \varepsilon u$ ;
- (iii)  $\varepsilon$  is the unique continuous kernel-valued function with that property;
- (iv) conversely, if  $u$  is a representative and  $\varepsilon : V \rightarrow K$  is continuous, then  $\varepsilon u$  is another representative of the same ordinary map.

The endpoints include `relFactor_mem_ker`, `relFactor_isKerFunOn`, `continuous_internal_reps_difference_unique_kerSign`, `isInternalRepOn_kerFun_mul`, and `internal_reps_free_transitive`. The phrase “free and transitive” is justified here at the level of the space of representatives under multiplication by continuous kernel-valued functions; no general torsor library is required.

For the certified projection, theorem 2 becomes

$$v = \varepsilon u, \quad \varepsilon : V \rightarrow \{\pm 1\}.$$

Thus all local freedom is reduced to one discrete two-valued function.

## 6 Discreteness turns ambiguity into local constancy

The topology of the kernel now becomes decisive. Since  $K$  is discrete, any continuous map into  $K$  is locally constant. Task 28 proves this for `toKerFun`; on a preconnected domain the function is constant.

**Corollary 1** (Connected-domain rigidity). *Let  $V$  be preconnected. If  $u$  and  $v$  are continuous representatives of the same ordinary map on  $V$ , then there is a single  $k \in K$  with*

$$v(x) = k u(x) \quad \forall x \in V.$$

*For the certified projection,  $k \in \{\pm 1\}$ .*

Task 28 also preserves the opposite statement. On a disconnected domain a continuous kernel-valued function need not be globally constant. The endpoint `exists_nonconstant_kerFun_of_disconnected` supplies an explicit negative control.

**Methodological principle 3** (One sign per component, not one sign per arbitrary overlap). The correct invariant object is a locally constant kernel-valued function. Replacing it by one global sign on an arbitrary disconnected overlap discards formally relevant information.

This repeats, at a higher layer, the lesson already encountered in earlier local-section analyses: connectedness assumptions control when discrete ambiguity collapses to one constant label.

## 7 Identity and inverse representatives: what is already normalized

Before entering triple overlaps, Task 28 checks the lower-dimensional transition identities. The constant identity map in  $\mathcal{L}$  represents the ordinary identity transition, and inversion preserves internal representability: if  $u_{ij}$  represents  $g_{ij}$ , then  $u_{ij}^{-1}$  represents  $g_{ji}$ .

If a representative  $u_{ji}$  is chosen independently, it need not equal  $u_{ij}^{-1}$ . The difference is again a unique continuous kernel-valued factor. This is the content of `inv_rep_difference_unique`.

These results are deliberately local. Task 28 does not impose a simultaneous global normalization of all identities and inverses across the refined system. That distinction will matter in the review of Level C below.

## 8 Hard target C: the triple-overlap defect

The decisive new object is derived on a common refined triple-overlap. Let  $u_{ij}$ ,  $u_{jk}$  and  $u_{ik}$  represent  $g_{ij}$ ,  $g_{jk}$  and  $g_{ik}$  respectively. Downstairs, the ordinary transition law is

$$g_{ik} = g_{jk}g_{ij}.$$

Therefore  $u_{jk}u_{ij}$  and  $u_{ik}$  are two internal representatives of the same ordinary map  $g_{ik}$ . The general ambiguity theorem can be applied immediately.

Task 28 defines

$$\delta_{ijk} := (u_{jk}u_{ij})u_{ik}^{-1}. \quad (3)$$

Equivalently,

$$u_{jk}u_{ij} = \delta_{ijk}u_{ik}. \quad (4)$$

**Theorem 3** (Triple-overlap defect). *On every common refined triple-overlap,  $\delta_{ijk}$  is:*

- (i) *kernel-valued;*
- (ii) *continuous;*
- (iii) *locally constant;*
- (iv) *uniquely determined by (4).*

*If the triple-overlap domain is preconnected,  $\delta_{ijk}$  is constant there.*

The principal endpoints are `tripleDefect_mem_ker`, `tripleDefect_isKerFunOn`, `tripleDefect_isLocallyConstant`, `tripleDefect_const_of_connected`, `triple_overlap_defect`, and `tripleDefect_unique`.

For the certified model,

$$\delta_{ijk} : V_{ijk} \longrightarrow \{\pm 1\}.$$

The formal derivation is important. The defect is not postulated as a new topological quantity and is not imported from an established obstruction theory. It is literally the relative factor between two internal representatives of the same ordinary transition.

## 9 Exact compatibility is equivalent to trivial defect

The central theorem boundary is unusually sharp.

**Theorem 4** (Exact compatibility criterion). *For a fixed chosen triple of internal representatives,*

$$u_{jk}u_{ij} = u_{ik} \iff \delta_{ijk} = 1$$

*pointwise on the common domain.*

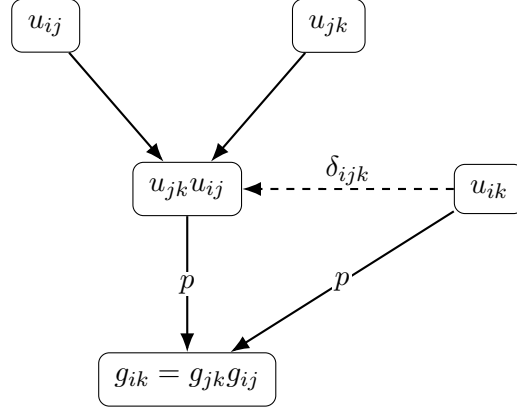


Figure 1: The defect is the unique kernel factor relating two internal representatives of the same ordinary transition. The dashed edge abbreviates  $u_{jk}u_{ij} = \delta_{ijk}u_{ik}$ .

This is the endpoint `exact_internal_transition_compatibility_iff_defect_trivial`. No existential conclusion is hidden in it. It does not say that representatives can be chosen with  $\delta_{ijk} = 1$ ; it says only that, for representatives already chosen, exact internal multiplication is equivalent to triviality of their derived defect.

That distinction is the fulcrum of Part XXVIII. Local continuous representatives are guaranteed, but exact global compatibility is no longer a local existence question.

## 10 Changing representatives and the exact defect law

Let the local representatives be changed through continuous kernel-valued functions,

$$u'_{ij} = \varepsilon_{ij}u_{ij}, \quad \varepsilon_{ij} : V_{ij} \rightarrow K. \quad (5)$$

Because  $K$  is central, the new triple defect can be rearranged explicitly. Task 28 proves the exact formula

$$\delta'_{ijk} = (\varepsilon_{jk}\varepsilon_{ij}\varepsilon_{ik}^{-1})\delta_{ijk}. \quad (6)$$

The ordering in (6) is not guessed. The source contains a separate algebraic lemma `central_rearrange` deriving the needed rearrangement from centrality and the inherited convention  $g_{ik} = g_{jk}g_{ij}$ .

**Corollary 2** (When the defect is unchanged). *Under (5), one has  $\delta'_{ijk} = \delta_{ijk}$  exactly when*

$$\varepsilon_{jk}\varepsilon_{ij} = \varepsilon_{ik}.$$

The endpoint is `defect_eq_iff_changeFactor_trivial`. Task 28 also proves by `defect_changes_of_nontrivial_kerFun` that allowed representative changes can genuinely alter the defect while preserving every ordinary transition downstairs.

This has an immediate methodological consequence: a concrete  $\delta_{ijk}$  attached to one arbitrary choice of representatives is not itself an absolute global invariant. The meaningful global question concerns whether the whole defect system can be made trivial by one simultaneous allowed change of representatives.

## 11 Associativity produces a fourfold consistency law

Although the Task-28 prompt treated the fourfold analysis as optional, Aristotle completed it. On a common refined fourfold overlap, associativity of multiplication in  $\mathcal{L}$  gives

$$\delta_{ijk}\delta_{ikl} = \delta_{jkl}\delta_{ijl}. \quad (7)$$

This is `quadruple_defect_consistency`.

The significance of (7) is structural. The triple defects do not form an arbitrary collection of locally constant kernel-valued functions; they themselves satisfy a compatibility law forced by associativity one level higher. Again, Task 28 does not assign an external name to this law and does not package the  $\delta$  data into a cohomology theory. The relation is obtained entirely from the internal group algebra already reconstructed.

**Methodological principle 4** (The next structure emerges from associativity). Once pairwise representative ambiguity is classified and triple defects are defined, associativity alone constrains the defect system on fourfold overlaps. No new physical or geometric input is required for this higher consistency relation.

## 12 Refinement does not erase the defect problem

Different choices of local section neighbourhoods produce different refinements. Task 28 therefore compares two representative systems on their common refinement. The two representatives differ by a unique continuous kernel-valued function, exactly as in theorem 2. Their triple defects are then related by the same transformation law (6).

The relevant endpoints are `refinement_comparison_kerFun` and `refinement_comparison_defect`. No direct limit over covers and no sheaf machinery is introduced.

This comparison is important for the next stage: the appearance of a defect is not an artefact of one particular refinement. Moving to a common refinement merely transports the ambiguity and defect through the already derived kernel transformation law.

## 13 The local action diagram reconnects to the frame geometry

Task 28 does not glue a lifted frame total space. It nevertheless reconnects each local internal representative to the one-fibre lifted-frame action from earlier work. Every representative projects exactly to the ordinary transition, and the local action diagram commutes.

Schematically,

$$\begin{array}{ccc} \mathcal{L} \times \widetilde{\mathcal{F}_0^+} & \longrightarrow & \widetilde{\mathcal{F}_0^+} \\ \downarrow p \times \pi & & \downarrow \pi \\ G_{\text{vis}} \times \mathcal{F}_0^+ & \longrightarrow & \mathcal{F}_0^+. \end{array}$$

Thus the new  $u_{ij}$  are not abstract lifts detached from the previous geometry: locally they act upstairs in a way that projects to exactly the ordinary frame-coordinate change downstairs. Task 28 stops before using these local actions as gluing maps for a total lifted space.

## 14 Levels A, B, and C: what has actually been closed

The endpoint module makes the progress hierarchy explicit.

| Level | Mathematical statement                                                                                                  | Status    |
|-------|-------------------------------------------------------------------------------------------------------------------------|-----------|
| A     | Every ordinary transition value has internal representatives; two pointwise representatives differ by a kernel element. | INHERITED |
| B     | After refining overlaps, continuous internal representatives exist.                                                     | PROVED    |
| C     | A candidate exists whose triple-overlap defects vanish everywhere.                                                      | OPEN      |

Level B is the principal positive result of Task 28. Level C is represented by

**LevelC**:  $\exists C : \text{InternalTransitionCandidate},$   
 $\text{IsCompatibleInternalTransitionSystem}(C).$

The latter predicate requires all stored triple defects to equal 1 on their common refined domains. Task 28 neither assumes nor proves existence of such a candidate.

The even sharper frontier is **CanTrivialiseSimultaneously**. For a fixed candidate  $C$ , it asks whether there exist continuous kernel-valued changes  $\varepsilon_{ij}$  that make all transformed triple defects trivial simultaneously. Aristotle proves only the forward consequence:

if such a simultaneous trivialization exists, then the modified representatives satisfy  
the exact ordinary triple-overlap multiplication law.

Existence remains open.

## 15 Review boundary: “defect-free” is not yet a fully glued transition system

The formal Task-28 result is strong, but one terminology choice in the audit deserves tightening before Part XXIX. The phrase “Level C = globally compatible internal transition system” is slightly stronger than the interface currently proves.

**IsCompatibleInternalTransitionSystemC** says precisely that every triple defect of the refined candidate vanishes. Consequently, exact triple multiplication holds on every common refined triple domain. However, Task 28 does not separately package the following consequences:

- (i) every stored representative of an identity transition is exactly the identity;
- (ii) every independently stored reverse representative equals the inverse of the forward representative;
- (iii) two refined representatives for the same ordered pair  $(i, j)$  agree exactly on their mutual intersection;
- (iv) the refined representatives therefore glue to one continuous map on the entire original overlap  $U_i \cap U_j$ .

Some of these statements are expected to follow quickly from defect-freeness by specializing the triple law to repeated indices, together with uniqueness and the local identity/inverse lemmas. They are nevertheless not Task-28 endpoints and must not be counted retrospectively as proved.

**Proof boundary 2** (Correct Level-C interpretation). *Task 28 proves a defect-free refined local candidate implies exact triple compatibility on the refined domains. It does not yet prove that such a candidate is identical to a single normalized, globally glued internal transition system on the original overlaps.*

This is not a repair of a false theorem. It identifies the first hardening obligation for Task 29.

## 16 The complete Task-28 mathematical ledger

| Claim                                                                       | Status             | Formal boundary                                                       |
|-----------------------------------------------------------------------------|--------------------|-----------------------------------------------------------------------|
| Every ordinary transition value is internally representable                 | inherited          | surjectivity of $p$                                                   |
| Every ordinary continuous transition is locally continuously representable  | proved             | local sections of $p$                                                 |
| A refined overlap cover carrying representatives exists                     | proved             | <code>OverlapRefinement</code>                                        |
| One continuous representative on every whole original overlap always exists | false/not<br>plied | im-<br>certified whole-domain counterexample<br>exact relative factor |
| Two representatives differ by a kernel-valued function                      | proved             |                                                                       |
| The difference is continuous and unique                                     | proved             | hard target B                                                         |
| The ambiguity is locally constant                                           | proved             | discreteness of $K$                                                   |
| One constant kernel element on connected domains                            | proved             | preconnectedness                                                      |
| One constant sign on arbitrary disconnected domains                         | refuted            | explicit negative control                                             |
| Triple defect $\delta_{ijk}$ is kernel-valued                               | proved             | derived from equal ordinary projection                                |
| Triple defect is continuous and locally constant                            | proved             | topological group + discrete kernel                                   |
| Exact triple compatibility iff $\delta_{ijk} = 1$                           | proved             | fixed representative system                                           |
| Representative-change law                                                   | proved             | centrality + inherited multiplication order                           |
| Defect is unchanged under every representative change                       | false              | explicit change theorem                                               |
| Fourfold consistency of defects                                             | proved             | associativity in $\mathcal{L}$                                        |
| Refinement comparison                                                       | proved             | common refinement + kernel factors                                    |
| Simultaneous trivialization of all defects                                  | open               | <code>CanTrivialiseSimultaneously</code>                              |
| A globally glued lifted total space                                         | not constructed    | outside Task 28                                                       |
| A spin structure                                                            | not constructed    | outside Task 28                                                       |

## 17 Preserved failed builds and engineering provenance

The Task-28 audit records twelve failed builds before the final green endpoint. None changed mathematical content. They are retained because they document where the formal proof scripts interacted nontrivially with the pinned Mathlib API.

| # | Location                                  | Failure diagnosis                                              | Repair                               |
|---|-------------------------------------------|----------------------------------------------------------------|--------------------------------------|
| 1 | <code>TransitionLiftBase</code>           | wrong application shape for <code>MonoidHom.mem_ker</code>     | remove placeholder argument          |
| 2 | same                                      | <code>field_simp</code> irrelevant to group cancellation       | use <code>mul_inv_eq_one</code>      |
| 3 | <code>LocalInternalRepresentatives</code> | syntactic mismatch in <code>Classical.choose_spec</code> chain | use stored <code>covers</code> field |

| #  | Location                    | Failure diagnosis                                                       | Repair                                             |
|----|-----------------------------|-------------------------------------------------------------------------|----------------------------------------------------|
| 4  | KernelAmbiguity             | rewrite targeted conjunction rather than continuity component           | use the first component explicitly                 |
| 5  | RepresentativeChannel       | over-eager cancellation in central rearrangement and quadruple identity | explicit <code>calc</code> chains using centrality |
| 6  | CertifiedProjection         | inherited namespace not opened                                          | add inherited <code>open</code> list               |
| 7  | same                        | continuity proof remained as a composition                              | beta-reduce with an explicit <code>show</code>     |
| 8  | same                        | <code>dite</code> under lambda not reduced                              | expose applied form before <code>dif_pos</code>    |
| 9  | InternalTransitionInterface | defect name hidden in namespace                                         | open <code>InternalProjection</code>               |
| 10 | ModelBridge                 | wrong theorem names for pinned Mathlib                                  | use revision-correct names                         |
| 11 | same                        | continuity had to be expressed through inherited spatial action         | prove entries by finite row cases                  |
| 12 | Task28 endpoint layer       | placement of <code>omit</code> before documentation string              | reorder syntax                                     |

No intended theorem was falsified during these repairs, and no counterexample replaced a target theorem.

## 18 Comparison with established mathematics

Only now, after the intrinsic construction has been completed, is it useful to compare the result with standard bundle and spin geometry. In the usual local description of a principal or frame bundle, transition functions  $g_{ij}$  satisfy the ordinary cocycle law on triple overlaps. Given a central double covering of structure groups, local lifts of  $g_{ij}$  need not satisfy the same law exactly; their failure is measured by central kernel elements on triple overlaps. Changing the local lifts changes these elements by the corresponding lower-overlap factors, while associativity imposes a higher overlap identity [8, 3, 4].

The Task-28 formulas have exactly this shape, but the project has reached them without importing that formalism:

$$\begin{aligned}
u_{jk}u_{ij} &= \delta_{ijk}u_{ik}, \\
\delta'_{ijk} &= (\varepsilon_{jk}\varepsilon_{ij}\varepsilon_{ik}^{-1})\delta_{ijk}, \\
\delta_{ijk}\delta_{ikl} &= \delta_{jkl}\delta_{ijl}.
\end{aligned}$$

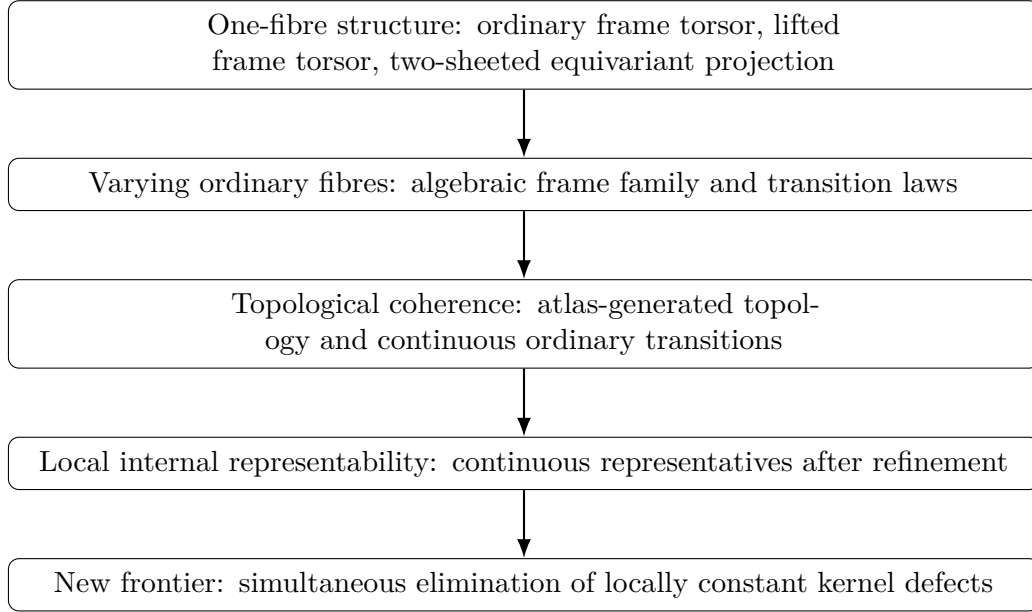
For the certified kernel  $K = \{\pm 1\}$ , the  $\delta$  system is therefore externally recognizable as the local data from which a  $\mathbb{Z}_2$ -valued degree-two obstruction is normally organized. In conventional treatments, the corresponding global obstruction for lifting an oriented orthonormal frame bundle through the standard double cover is represented by the second Stiefel–Whitney class  $w_2$ ; its vanishing is the existence criterion for a spin structure [5, 4].

**Proof boundary 3** (External comparison is not an internal theorem). *Task 28 does not define a Čech complex, a cohomology class,  $w_2$ , or a spin structure. It proves the raw local ingredients and their transformation laws. The statement that these data match the standard local pattern leading to  $w_2$  is an external mathematical comparison, not a Lean theorem of the current project.*

This distinction is especially important because a concrete defect function depends on the chosen representatives. Established obstruction theory passes from these representatives to an equivalence class under exactly the type of change law now reconstructed. That quotient step has not yet been formalized here. The project is therefore at the threshold of the conventional obstruction layer, not beyond it.

## 19 Conservation of Difficulty

The trajectory from Parts XXV–XXVIII exhibits a particularly clean relocation of difficulty.



The first three stages progressively removed missing structure. Task 28 is different: it shows that, once local existence is guaranteed, the remaining problem becomes a system of exact compatibility equations. The difficulty is no longer hidden in topology of the group projection, because local sections already solve that. It is no longer hidden in ordinary frame topology, because Task 27 solved that. It now sits in whether the discrete kernel discrepancies can be removed *simultaneously* over all overlaps.

**Methodological principle 5** (Difficulty has become global synchronization). Every local transition can be represented continuously. Every pair of representatives is completely classified. Every triple failure is measured by a locally constant kernel element with a known transformation law. What remains is the global synchronization problem: can one choose all local representatives at once so that all these failures vanish together?

That is a much narrower question than the original search for a spin-like structure, but it is also the mathematically decisive one.

## 20 Part XXIX: the next task

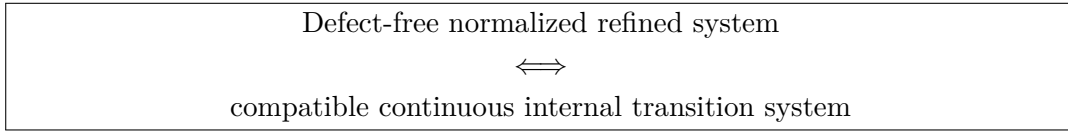
Part XXIX should not begin by importing a cohomological answer. The formal project already contains enough intrinsic data to state the next problem directly. The next task should have two tightly controlled phases.

## 20.1 Phase I: harden the meaning of defect-free

Before solving the global existence problem, Task 29 should close the review boundary of section 15. Starting from a defect-free refined candidate, it should prove or refute:

- (1) identity representatives normalize to 1;
- (2) reverse representatives normalize to inverses;
- (3) two refined representatives for the same ordered pair agree on common subdomains;
- (4) these representatives glue to a single continuous map on the entire original overlap;
- (5) the glued maps satisfy all identity, inverse, and triple laws exactly.

The desired endpoint is a genuine equivalence:



If any implication fails, the failure must be preserved and the correct additional datum isolated.

## 20.2 Phase II: simultaneous trivialization

Only after Phase I should Task 29 attack the existing predicate `CanTrivialiseSimultaneously`. With

$$\delta'_{ijk} = (\varepsilon_{jk}\varepsilon_{ij}\varepsilon_{ik}^{-1})\delta_{ijk},$$

the direct question is:

Do there exist continuous kernel-valued functions  $\varepsilon_{ij}$  on the refined overlaps such that every transformed defect  $\delta'_{ijk}$  equals 1 simultaneously?

The task should derive necessary and sufficient conditions from the already proved change law and quadruple consistency before any comparison with standard obstruction theory is introduced.

Three outcomes are mathematically meaningful:

- (i) the system is always simultaneously trivializable under the current hypotheses;
- (ii) explicit counterexamples exist, revealing a genuine global obstruction;
- (iii) additional hypotheses on the base/cover are needed, and the exact missing condition can be isolated.

Task 29 should stop at the strongest of these outcomes that can be proved formally.

## 20.3 What should remain for a later task

Even if simultaneous trivialization is solved positively for a given system, Part XXIX should not yet identify the result with a spin structure. A subsequent task would still need to glue the local products using the compatible internal transitions, construct the global total space, prove the free internal-group action, and prove the two-to-one equivariant map to the ordinary frame total space. Only after that global object exists should the project perform the final comparison with the established definition.

## 21 Conclusion

Part XXVIII closes the local continuous representation problem and, in doing so, exposes the first nontrivial global compatibility system in exact form. Every ordinary transition is locally representable through the internal projection after refinement. All freedom between two such representatives is a unique continuous kernel-valued function; for the certified project it is a locally constant  $\{\pm 1\}$  function. On triple overlaps, internal multiplication closes only up to the derived defect

$$\delta_{ijk} = (u_{jk}u_{ij})u_{ik}^{-1} \in \{\pm 1\},$$

which is continuous, locally constant, and trivial exactly when the chosen internal transitions satisfy the ordinary multiplication law exactly. Allowed changes of representatives transform this defect by

$$\delta'_{ijk} = (\varepsilon_{jk}\varepsilon_{ij}\varepsilon_{ik}^{-1})\delta_{ijk},$$

and associativity forces the fourfold law

$$\delta_{ijk}\delta_{ikl} = \delta_{jkl}\delta_{ijl}.$$

The project has therefore crossed an important threshold. The remaining difficulty is no longer an unidentified missing object. It is a precisely formulated simultaneous compatibility problem for a discrete central kernel. Yet the final global structure is still absent, and the current Level-C predicate is not retrospectively stronger than its proved content. The next task must first convert defect-freeness into a fully glued normalized transition system and then determine whether all defects can be removed at once. Only after those questions are settled will the reconstruction have reached the point where a global lifted total space can be built and compared formally with the established notion of a spin structure.

**Status 1** (End of Part XXVIII). **Proved:** local continuous internal representatives after refinement; unique continuous kernel ambiguity; local constancy; intrinsic triple-overlap defect; exact compatibility iff trivial defect; exact representative-change law; fourfold defect consistency; refinement comparison; local action compatibility.

**Open:** simultaneous trivialization of all defects; normalization/gluing of a defect-free refined system into one transition system on original overlaps; construction of a lifted total space; identification with a spin structure.

**Next formal frontier:** harden defect-free compatibility and solve or delimit `CanTrivialiseSimultaneously`.

## References

- [1] Aristotle (Harmonic). *Task 27: Topological Coherence of the Varying Frame Family*. Formal Lean archive supplied 5 September 2026; focused build 8237 jobs; whole-project build 8308 jobs; SHA-256 df46fc45474acaa8b015c0ff9f96f4018876076c2ce97b60fe18a8bcb6f933a. 2026.
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- [8] Norman Steenrod. *The Topology of Fibre Bundles*. Princeton University Press, 1951.